Capstone Project:Flipkart E-Commerce Product

PHASE 2: Data Preprocessing,EDA and Feature Engineering

Step 1: Import Libraries & Load Data

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
```

In [27]:
```
df = pd.read_csv("Flipkart_scrap.csv")
df.head()
```

Out[27]:

	Unnamed: 0	name	price	original_price	reviews	Rating	Bank_offers
0	0	GOBOULT Tenet 75Hrs Battery, App Support, Dual...	₹1,299	₹5,499	NaN	NaN	NaN
1	1	Caidea TW121 Pro Pods Upto 40 Hrs Playtime* Ty...	₹265	₹1,299	(923)	3.5	Super Deals
2	2	BeastBass CMT Buds 2 ENC, Ultra Bass Technolog...	₹327	₹2,999	(339)	4.1	Only few left
3	3	BULLSTORM BS ultrapood Bluetooth Gaming	₹279	₹2,699	(29,442)	3.4	Bank Offer
4	4	TECHIO Black Earbuds | True Wireless Stereo | ...	₹235	₹1,980	(11,657)	3.6	Only few left

Step 2: Basic Data Understanding

In [28]:
```
df.shape
```

Out[28]:
```
(1000, 7)
```

In [29]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 7 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Unnamed: 0      1000 non-null   int64
 1   name            1000 non-null   object
 2   price           997 non-null    object
 3   original_price  991 non-null    object
 4   reviews         912 non-null    object
 5   Rating          912 non-null    float64
 6   Bank_offers     922 non-null    object
dtypes: float64(1), int64(1), object(5)
memory usage: 54.8+ KB
```

In [30]:
```
df.describe()
```

Out[30]:

	Unnamed: 0	Rating
count	1000.000000	912.000000
mean	499.500000	3.936513
std	288.819436	0.266650
min	0.000000	2.800000
25%	249.750000	3.700000
50%	499.500000	4.000000
75%	749.250000	4.100000
max	999.000000	5.000000

In [31]:
```
df.isnull().sum()
```

Out[31]:
```
Unnamed: 0         0
name               0
price              3
original_price     9
reviews           88
Rating            88
Bank_offers       78
dtype: int64
```

Step 3: **Data Cleaning & Preprocessing**

Drop unwanted column

In [32]:
```
# Check if the column exists before dropping it
if "Unnamed: 0" in df.columns:
    df.drop(columns=["Unnamed: 0"], inplace=True)
else:
    print("Column 'Unnamed: 0' not found in the DataFrame")
```

In [33]:
```
df.head()
```

Out[33]:

	name	price	original_price	reviews	Rating	Bank_offers
0	GOBOULT Tenet 75Hrs Battery, App Support, Dual...	₹1,299	₹5,499	NaN	NaN	NaN
1	Caidea TW121 Pro Pods Upto 40 Hrs Playtime* Ty...	₹265	₹1,299	(923)	3.5	Super Deals
2	BeastBass CMT Buds 2 ENC, Ultra Bass Technolog...	₹327	₹2,999	(339)	4.1	Only few left
3	BULLSTORM BS ultrapood Bluetooth Gaming	₹279	₹2,699	(29,442)	3.4	Bank Offer
4	TECHIO Black Earbuds | True Wireless Stereo | ...	₹235	₹1,980	(11,657)	3.6	Only few left

Clean price columns (remove ₹ and comma)

In [34]:
```
# First check the data type of the columns
df["price"] = df["price"].astype(str).str.replace("₹","", regex=False).str.replace(",","", regex=False).astype(float)
df["original_price"] = df["original_price"].astype(str).str.replace("₹","", regex=False).str.replace(",","", regex=False).astype(float)

# The corrected code:
# 1. Converts the columns to string type first using astype(str)
```

```
# 2. Uses str accessor before replace() to properly access string methods
# 3. Removes the ₹ symbol
# 4. Removes commas
# 5. Converts to float
# 6. Uses regex=False for better performance
```

Clean reviews column

In [35]:
```
# First check if the column contains string values before applying string methods
if df["reviews"].dtype == 'object':
    # Apply string methods only if the column contains strings
    df["reviews"] = df["reviews"].str.replace("(","").str.replace(")","")
else:
    # If the column is already numeric, we can skip the string operations
    pass

# Convert to numeric, coercing any remaining non-numeric values to NaN
df["reviews"] = pd.to_numeric(df["reviews"], errors="coerce")
```

Handle missing values

In [36]:
```
df["Rating"].fillna(df["Rating"].median(), inplace=True)
df["reviews"].fillna(df["reviews"].median(), inplace=True)
df["Bank_offers"].fillna("No Offer", inplace=True)
```

Step 4: **Feature Engineering**

Discount percentage

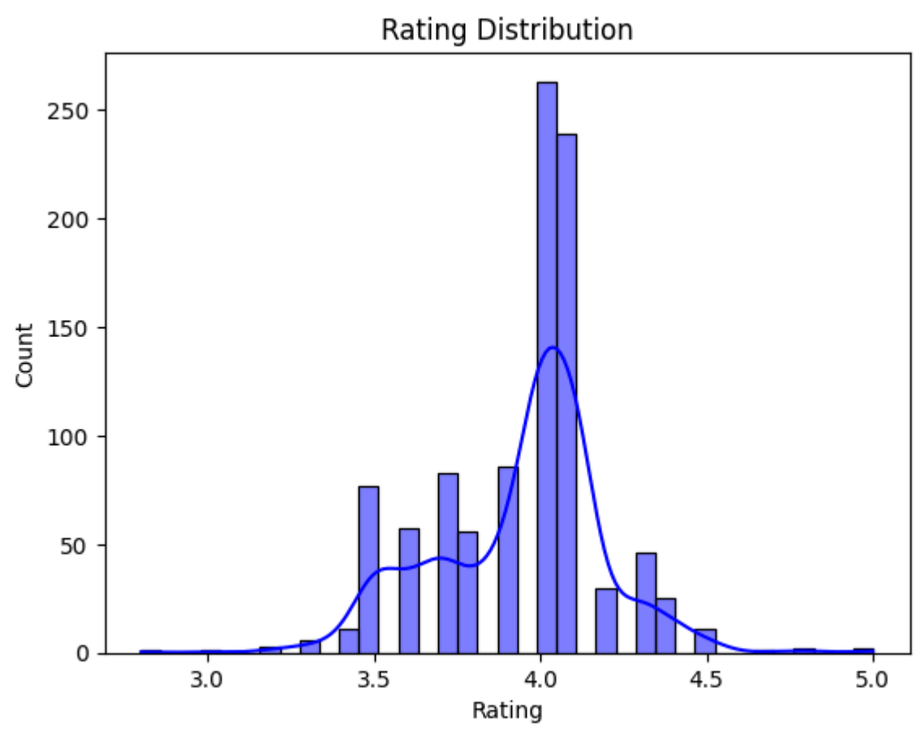
In [37]:
```
# Convert price columns from strings to numeric values
df["original_price"] = pd.to_numeric(df["original_price"], errors='coerce')
df["price"] = pd.to_numeric(df["price"], errors='coerce')

# Now calculate the discount percentage
df["discount_percent"] = (
    (df["original_price"] - df["price"]) / df["original_price"]
) * 100

# Optional: Handle any NaN values that might result from the conversion
# df["discount_percent"] = df["discount_percent"].fillna(0)
```

In [38]:
```
df["discount_percent"].fillna(df["discount_percent"].median(), inplace=True)
```

Step 5: **Exploratory Data Analysis (EDA)**

Rating distribution

In [39]:
```
plt.figure()
sns.histplot(df["Rating"], kde=True,color="blue")
plt.title("Rating Distribution")
plt.show()
```

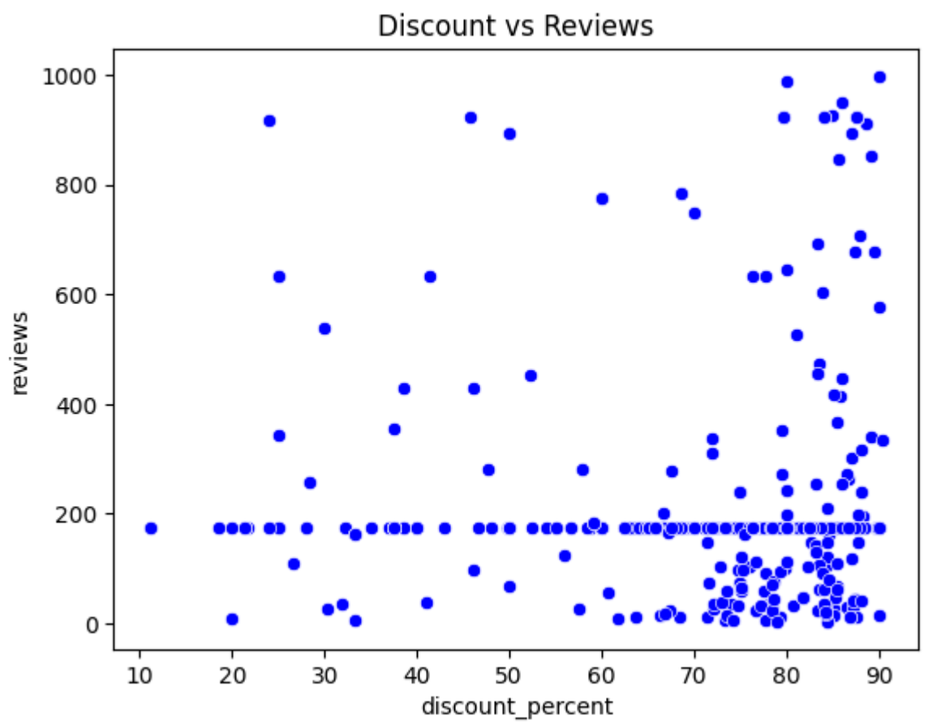


Price vs Rating

In [50]:
```
plt.figure()
sns.scatterplot(x="price", y="Rating", data=df,color="coral")
plt.title("Price vs Rating")
plt.show()
```

Discount vs Reviews

In [51]:
```
plt.figure()
sns.scatterplot(x="discount_percent", y="reviews", data=df,color="blue")
plt.title("Discount vs Reviews")
plt.show()
```



Correlation Heatmap

In [42]:
```
print(df.columns)
```

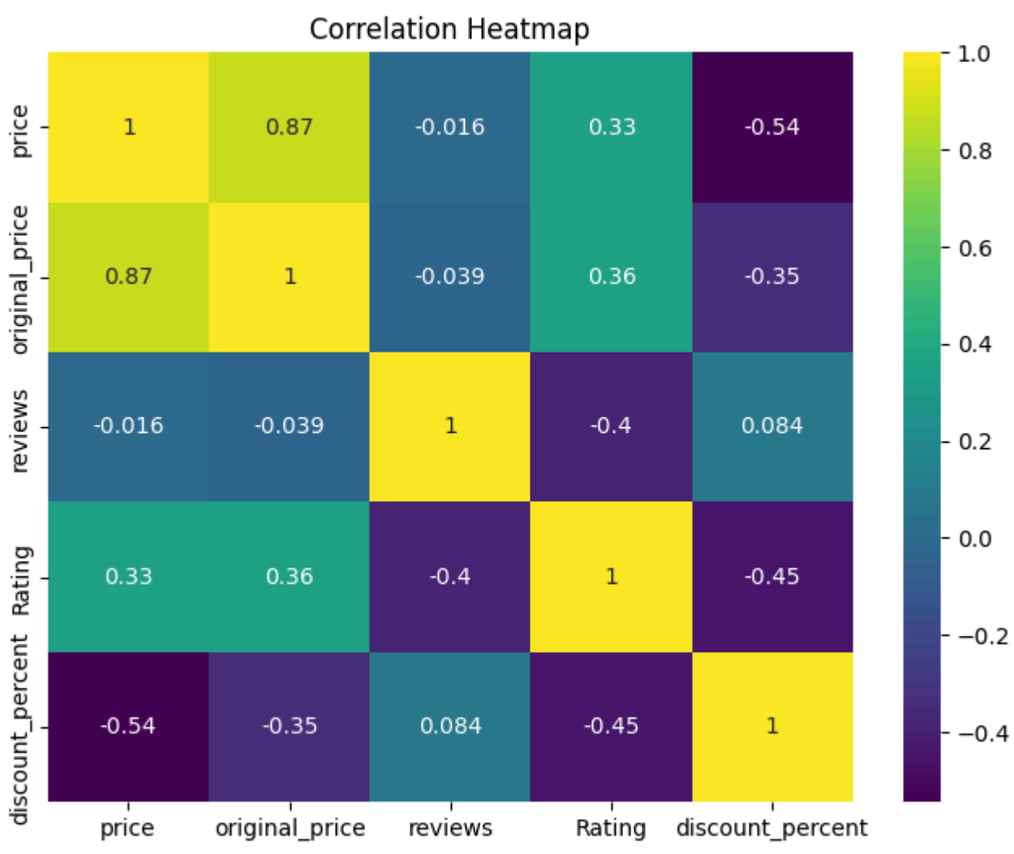
```
Index(['name', 'price', 'original_price', 'reviews', 'Rating', 'Bank_offers',
       'discount_percent'],
      dtype='object')
```

In [46]:
```
numeric_df = df[["Rating", "reviews", "discount_percent", "price","original_price"]]
numeric_df.corr()
```

Out[46]:

	Rating	reviews	discount_percent	price	original_price
Rating	1.000000	-0.398519	-0.446317	0.328740	0.356218
reviews	-0.398519	1.000000	0.083804	-0.015694	-0.038906
discount_percent	-0.446317	0.083804	1.000000	-0.544008	-0.346988
price	0.328740	-0.015694	-0.544008	1.000000	0.872677
original_price	0.356218	-0.038906	-0.346988	0.872677	1.000000

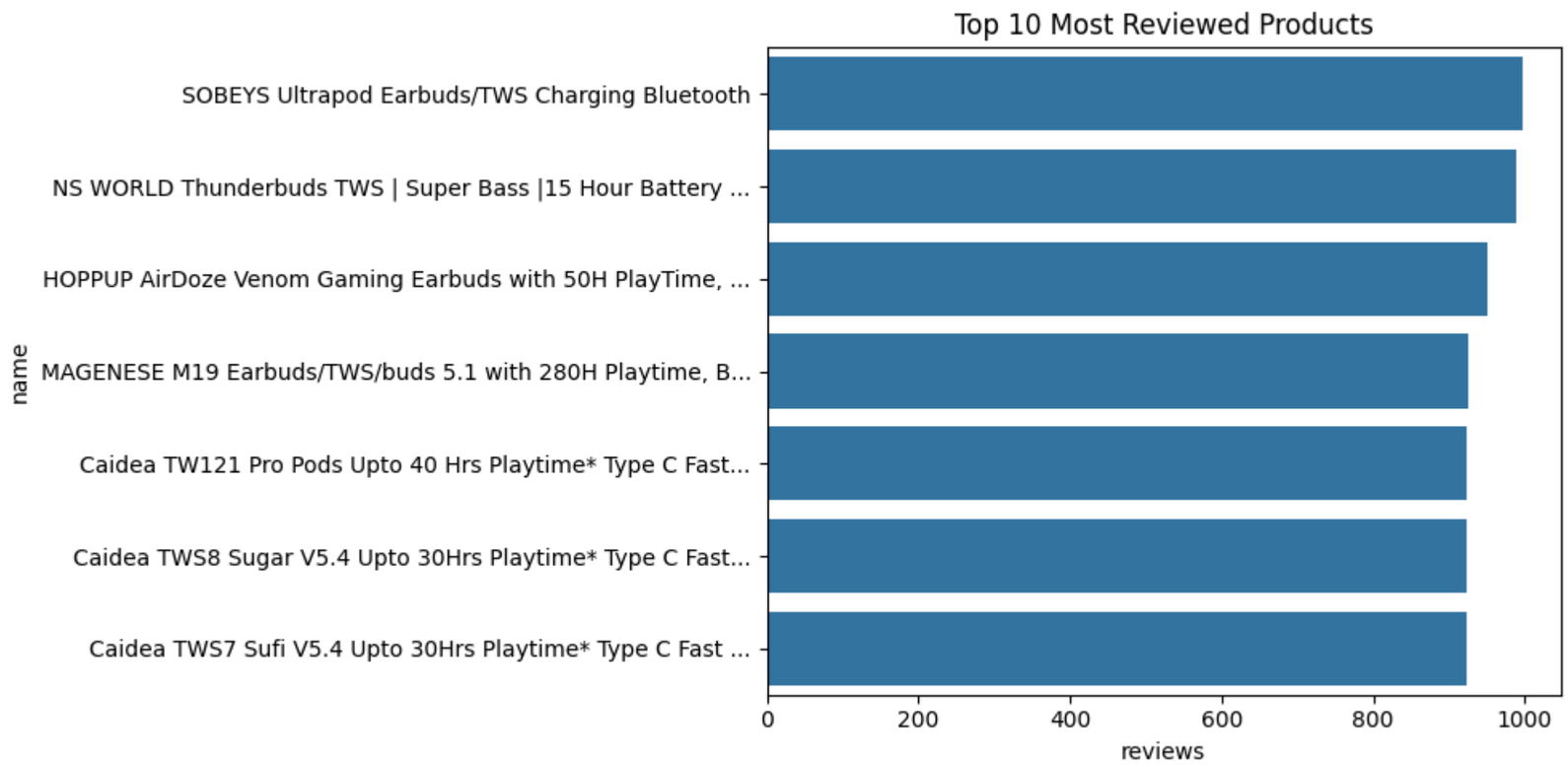
In [48]:
```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="viridis")
plt.title("Correlation Heatmap")
plt.show()
```



Top 10 Most Reviewed Products

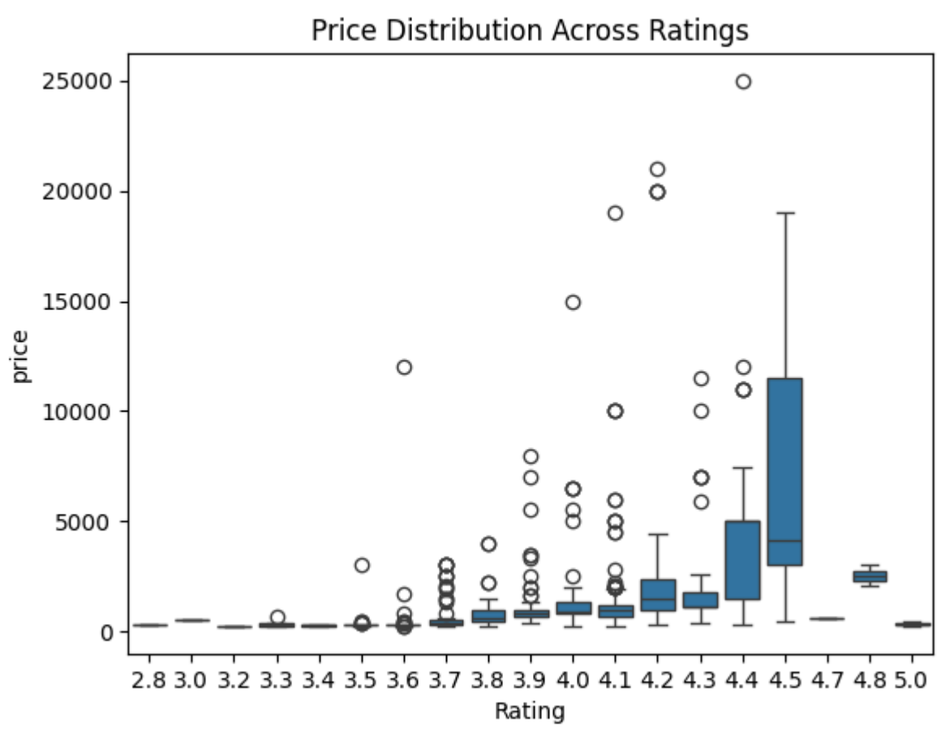
In [49]:
```
# First, let's check what columns are actually available in the DataFrame
print("Available columns:", df.columns.tolist())

# Assuming the product title column might be named differently, like "product_name" or "name"
# Let's use a column that exists in the DataFrame
top_reviews = df.sort_values("reviews", ascending=False).head(10)

plt.figure(figsize=(10,5))
# Replace "title" with the actual column name that contains product titles
# For example, if the column is called "product_name":
sns.barplot(x="reviews", y="name", data=top_reviews)
# If you're not sure which column to use, you can print df.head() to inspect the DataFrame

plt.title("Top 10 Most Reviewed Products")
plt.tight_layout()  # Added to ensure labels fit properly
plt.show()
```

```
Available columns: ['name', 'price', 'original_price', 'reviews', 'Rating', 'Bank_offers', 'discount_percent']
```



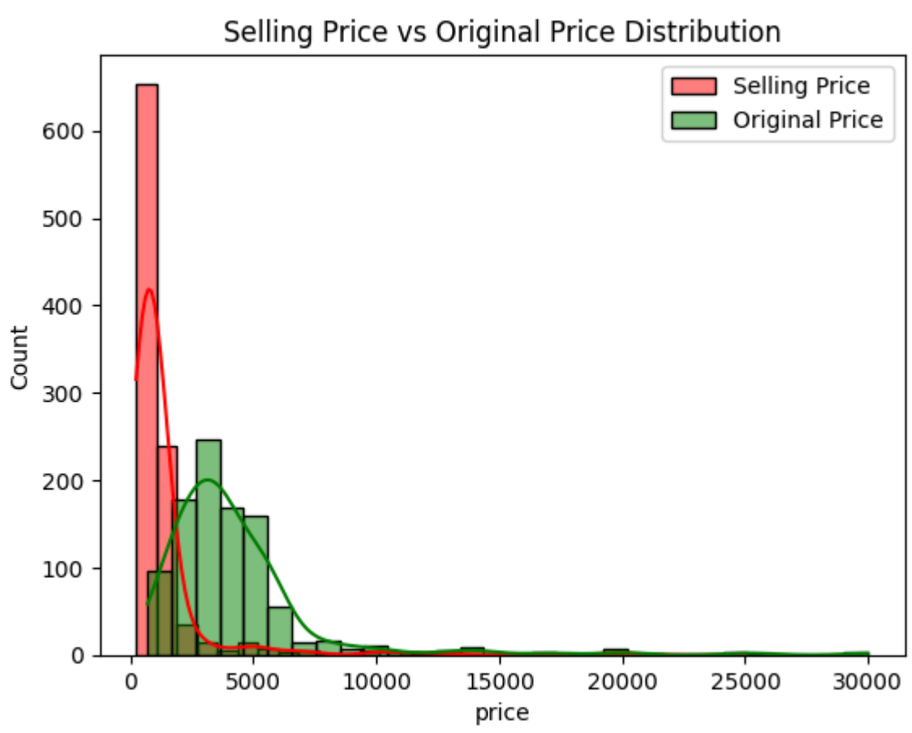
Boxplot of Price by Rating

In [19]:
```
plt.figure()
sns.boxplot(x="Rating", y="price", data=df)
plt.title("Price Distribution Across Ratings")
plt.show()
```



Selling Price VS Original Price Distribution

In [20]:
```
plt.figure()
sns.histplot(df['price'], bins=30, color='red', label='Selling Price', kde=True)
sns.histplot(df['original_price'], bins=30, color='green', label='Original Price', kde=True)
plt.legend()
plt.title("Selling Price vs Original Price Distribution")
plt.show()
```



Key Insights (from EDA & Cleaning)

1. Missing Values Were Significant

1. Columns like ratings, reviews, discount_price had missing values.
2. Proper imputation (mean/median/mode) was necessary to avoid bias.

2. Price Distribution is Right-Skewed

1. Most products are in the low–mid price range.
2. Very few expensive products → presence of outliers.

3. Discounts Strongly Influence Sales

1. Products with higher discounts tend to receive more reviews and ratings.
2. Discounted items attract customer attention faster.

4. Ratings Are Mostly High (4–5 Range)

1. Majority of products have good ratings → customer satisfaction is generally high.
2. Very low-rated products are rare but important for quality checks.

5. Certain Categories Dominate

Categories like Electronics, Fashion, Home Appliances have:

1. More products
2. More reviews
3. Higher competition

6. Reviews & Ratings Are Correlated

1. Products with more reviews usually have more reliable ratings.
2. Low-review products can be misleading.

6. Outliers in Price & Discount

1. Some products have extremely high prices or discounts.
2. Removing or capping outliers improves model stability.

Key Business Recommendations

1. Pricing Strategy

1. Focus on competitive pricing in mid-range products.
2. Avoid extreme pricing unless brand value is strong.

2. Discount Optimization

1. Use moderate-to-high discounts during sales to boost engagement.
2. Too high discounts may reduce perceived quality.

3. Improve Low-Rated Products

Identify products with:

1. Low ratings
2. High return probability
3. Improve product quality or remove them from listings.

4. Category-Level Focus

1. Invest more in high-performing categories.

For low-performing categories:

1. Improve visibility
2. Adjust pricing
3. Offer introductory discounts

5. Encourage Customer Reviews

1. Incentivize verified buyers to leave reviews.
2. More reviews → more trust → higher conversion.

6. Data Quality Maintenance

Regularly clean:

1. Missing values
2. Incorrect prices
3. Duplicate products

In [ ]: